

Networks, topologies and architecture ★ Golden

One college period · 40–45 minutes · Computer Communication & Networks

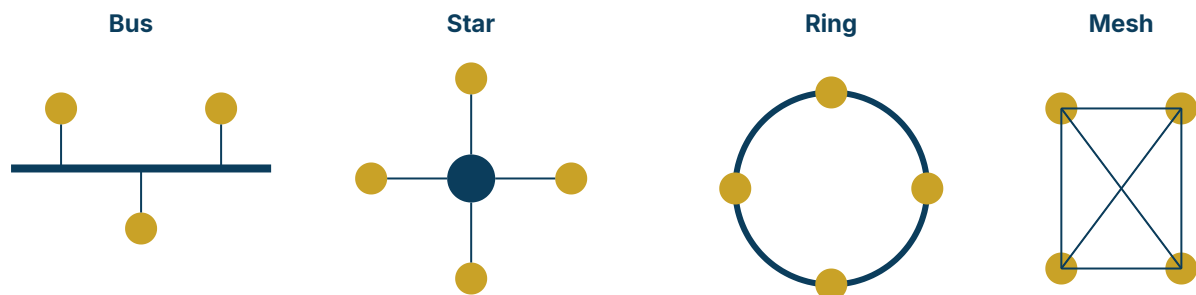
What you will learn

- Define network and its advantages
- Classify LAN, MAN, WAN
- Draw bus, star, ring, mesh, tree, hybrid topologies
- Name ISO, IEEE, ITU; introduce OSI and TCP/IP

Computer network. Two or more computers connected to share resources (files, printers, internet) and communicate.

Advantages: resource sharing, cheaper software/hardware per user, easy communication, central backup. Disadvantages: security risk, server failure, cabling cost, viruses spread faster.

Type	Span	Example
LAN	Room, lab, building	College computer lab
MAN	City	Cable TV / city Wi-Fi backbone
WAN	Country / world	Internet, bank branches



Four topologies every Section C diagram should match.

Topology	How linked	Plus / minus
Bus	One backbone cable	Cheapest; cable break kills the LAN; BIEK MCQ: cheapest topology
Star	Each node to a hub/switch	Easy to add PCs; centre fails ⇒ all fail
Ring	Circle, token passing (old)	Equal access; one break can stop ring
Mesh	Many-to-many	Very reliable; expensive
Tree	Stars of stars	Scalable campus
Hybrid	Mix	Real offices

Standard bodies: ISO (OSI model), IEEE (802.3 Ethernet, 802.11 Wi-Fi), ITU (telecom), ASCII (character codes — often listed with them in the Sindh book).

HOW BIEK ASKS THIS

Long 2026 OR: "What is meant by Computer Network? Classify by their scale and limitation." Write LAN/MAN/WAN with distance and one limitation each, then topologies with diagrams.

اردو میں سمجھیں

نیٹ ورک دو یا زیادہ کمپیوٹر جو وسائل بانٹتے ہیں۔ LAN عمارت، MAN شہر، WAN دنیا (انٹرنیٹ)۔
بس سب سے سستی، اسٹار سوئچ پر، رینگ دائرہ، میش ہر ایک سے ہر ایک۔

CHECK YOURSELF

1. Internet is which scale of network?
2. Cheapest topology?

Answers: 1. WAN · 2. Bus

MCQs

1. The cheapest topology is:
 - a. Ring
 - b. Bus
 - c. Tree
 - d. Star
2. OSI model contains:
 - a. six layers
 - b. five layers
 - c. three layers
 - d. seven layers
3. Which is NOT a type of computer network?
 - a. LAN
 - b. MAN
 - c. WAN
 - d. CPU

Key: 1=b, 2=d, 3=d

Short questions (3 marks each)

1. What is a computer network? Give advantages.
2. Differentiate LAN and WAN.
3. Write short notes on star and bus topology.

Model short answers

1. Linked computers sharing resources. Advantages: sharing, communication, backup, cost.
2. LAN is local and fast (lab). WAN covers large distance (internet) and is slower/more expensive.
3. Star: nodes to central switch, easy management. Bus: shared cable, cheap, collision and break problems.

Long questions (board Section C)

1. What is meant by a computer network? Classify by scale and explain topologies with diagrams.